Akshat Kankani

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Education

Manipal Institute of Technology — Bengaluru, KA
B.Tech in Computer Science (Artificial Intelligence) — CGPA: 8.11 — *July 2023 – May 2027*

Coursework: *Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, Machine Learning, DevOps & Cloud Computing*

Projects

Strata | *C++, POSIX APIs (file I/O, fsync, signals), Multithreading, Make - Team Project (2 Members)* — GitHub

- Built a **C++ time-series storage engine** with **write-ahead logging**, crash recovery and LSM-style compaction.
- Implemented **Gorilla-based compression** using delta-of-delta and XOR encoding, reducing storage by **4.28×**.
- Built an **inverted index** for high-cardinality queries with **sub-microsecond p50 latency** across 1M series.
- Proved crash safety via **deterministic fault injection**, killing the process at 7 exact code points mid-write/compaction with **zero data loss**.

Low-Latency Matching Engine | *C++, CMake, GoogleTest, Hardware Performance Profiling* — GitHub

- Built a **C++ limit-order-book matching engine** with **price-time priority**, partial fills, and O(1) order cancellation.
- Tested four data-structure designs, improving throughput **2.5–3.9× across three workloads (up to 14.5M ops/sec)**.
- Profiled hardware performance counters to identify a branch-misprediction bottleneck, reducing its cost by over a third and **improving throughput by up to 122%** with an improved architecture.

Relay | *Python, FastAPI, SQLAlchemy, PostgreSQL/pgvector, RAG, Redis, Celery* — Live Demo | GitHub

- Built an **LLM-orchestrated context engine** connecting GitHub, Slack, and Jira across 8 cross-system query modes, to help teams find code, investigate issues, and capture engineering knowledge.
- Implemented **hybrid keyword/vector retrieval** and **relevance scoring**, separating true matches from noisy correlations across integrated systems.
- Designed asynchronous **RAG pipelines** using Celery/Redis to handle LLM processing at scale, and integrated four LLM providers, backed by 300+ tests.

PRScope | *Python, SAST, FastAPI, PostgreSQL, ChromaDB, GitHub Actions* — Live Demo | GitHub

- Built a Chrome extension that auto-reviews and evaluates GitHub PRs inline, flagging **security vulnerabilities, architecture violations, and cross-file breakage** before merge.
- Architected a **risk-scoring** engine that maps how a change ripples across a codebase, surfacing callers and dependencies the diff itself never touches.
- Developed an **incident-retrieval** feature that matches PRs against 15 real production outages (Log4Shell, GitLab, Knight Capital), validated at 93% precision@1.

Technical Skills

Languages: C++, Python, SQL
Databases: PostgreSQL, MongoDB, Redis
Backend & Infrastructure: FastAPI, REST APIs, Docker, Kubernetes, AWS, Celery, SQLAlchemy
Developer Tools: Git, GitHub, Linux, Pytest, GoogleTest, GitHub Actions (CI/CD), Postman
Systems: System Design, Concurrency & Multithreading, Performance Optimization, Software Testing
AI & ML: NumPy, Pandas, scikit-learn, LLMs, RAG, MCP

Achievements & Certifications

- Solved **300+** Data Structures & Algorithms problems in **C++** across LeetCode and other coding platforms.
- Achieved a peak **LeetCode Contest Rating of 1862** (**Knight**) through consistent participation in coding contests.
- Certified Kubernetes Application Developer (CKAD) Coursera — [Pearson]